MAT092: Remedial Course in Mathematics

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LECTURE-02

From Relations to Functions

Motivation: Not all relations behave predictably. Sometimes, we want every input to produce exactly one output — no ambiguity.

Definition: A function f from a set A to a set B is a relation $f \subseteq A \times B$ such that

$$\forall a \in A,\ \exists\, b \in B \text{ with } (a, b) \in f\,.$$

Notation: We write $b = f(a)$, and denote the function as

$$f : A \to B.$$

Intuition: A function is like a machine — every input goes in once, every output comes out exactly once.

Example:

$$f(x) = x^2, \quad A = \mathbb{R},\ B = \mathbb{R}.$$

Every real x gives a single output x^2.

Domain, Codomain, and Range

Key Components:

Domain: Set of all admissible inputs.

Codomain: Set in which outputs live.

Range (Image): Actual outputs that appear.

Example:

$$f : \mathbb{R} \to \mathbb{R},\ f(x) = x^2.$$

Domain $= \mathbb{R}$, Codomain $= \mathbb{R}$, Range $= [0, \infty)$.

Remark: Changing the codomain changes the function! For instance,

$$f : \mathbb{R} \to [0, \infty) \quad \text{is not the same as} \quad f : \mathbb{R} \to \mathbb{R}.$$

Visualizing a Function

Graphical Idea: The graph of a function $f : A \to B$ is

$$G(f) = \{(x, f(x)) : x \in A\} \subseteq A \times B.$$

Vertical Line Test: A curve in the plane represents a function if every vertical line cuts it at most once.

Example:

$y = x^2 \Rightarrow$ a parabola $\to$ passes the vertical line test.

$x^2 + y^2 = 1 \Rightarrow$ a circle $\to$ fails the test.

Intuition: The vertical line test encodes the idea "each x gives one y".

Properties of Functions

1. Injective (One-to-one): Distinct inputs give distinct outputs.

$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

Example: $f(x) = 2x + 1$ is injective.

2. Surjective (Onto): Every element of the codomain is hit by some input.

$$\forall y \in B,\ \exists x \in A \text{ such that } f(x) = y.$$

Example: $f(x) = x^3 : \mathbb{R} \to \mathbb{R}$ is surjective.

3. Bijective: Both injective and surjective. Such functions are invertible.
Example: $f(x) = 2x + 3$ has inverse $f^{-1}(y) = \frac{y-3}{2}$.

Composition and Inverse of Functions

Composition: Given $f : A \to B$ and $g : B \to C$, define

$$(g \circ f)(x) = g(f(x)).$$

Associative but not necessarily commutative.

Example:

$$f(x) = 2x, \quad g(x) = x + 3.$$

Then

$$(g \circ f)(x) = 2x + 3, \quad (f \circ g)(x) = 2(x + 3) = 2x + 6.$$

Inverse Function: If f is bijective, there exists $f^{-1} : B \to A$ such that

$$f^{-1}(f(x)) = x \text{ and } f(f^{-1}(y)) = y.$$

Understanding Injectivity and Surjectivity

Recap of Definitions:

$f : A \to B$ is **injective** (one-to-one) if

$$f(x_1) = f(x_2) \implies x_1 = x_2.$$

$f : A \to B$ is **surjective** (onto) if

$$\forall y \in B,\ \exists x \in A \text{ such that } f(x) = y.$$

Why domain and codomain matter:

Injectivity depends on how different inputs behave.

Surjectivity depends on whether all possible outputs (in the codomain) are reached.

Moral: The same rule $f(x)$ can be injective or not, surjective or not — depending on the chosen sets A and B.

Example 1: $f(x) = x^2$ — Not Always Injective

Case 1: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$

$$f(2) = 4 = f(-2), \quad 2 \neq -2.$$

Hence, f is **not injective** on $\mathbb{R}$.

Case 2: $f : [0, \infty) \to \mathbb{R}$, $f(x) = x^2$
Now if $f(x_1) = f(x_2)$ then $x_1^2 = x_2^2 \implies x_1 = x_2$ (since both nonnegative).
Hence: f is injective on $[0, \infty)$.

Lesson: Injectivity can depend entirely on the chosen domain.

Example 2: $f(x) = x^2$ — Surjective or Not?

Case 1: $f : \mathbb{R} \to \mathbb{R}$
The range is $[0, \infty)$, but codomain is $\mathbb{R}$.

$$\Rightarrow f \text{ is not surjective because no } x \text{ gives } f(x) = -1.$$

Case 2: $f : \mathbb{R} \to [0, \infty)$
Now for every $y \geq 0$, there exists $x = \pm\sqrt{y}$ such that $f(x) = y$.

$$\Rightarrow f \text{ is surjective.}$$

Lesson: Surjectivity depends on the choice of codomain.

Example 3: $f(x) = 2x + 3$ — A Perfectly Bijective Function

Function: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + 3$
Injective: If $f(x_1) = f(x_2)$ then $2x_1 + 3 = 2x_2 + 3 \Rightarrow x_1 = x_2$.
Surjective: Given any $y \in \mathbb{R}$, choose $x = \frac{y-3}{2}$ so that $f(x) = y$.
Hence: f is both injective and surjective — a bijection.

Inverse:

$$f^{-1}(y) = \frac{y-3}{2}.$$

Intuition: Linear functions with nonzero slope are always bijective from $\mathbb{R} \to \mathbb{R}$.

Example 4: Trigonometric Function and Domain Restriction

Function: $f : \mathbb{R} \to [-1, 1]$, $f(x) = \sin x$

Not injective on $\mathbb{R}$ since $\sin x = \sin(x + 2\pi n)$.

Surjective onto $[-1, 1]$ because every value in $[-1, 1]$ is achieved.

Now restrict domain: $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1, 1]$

Injective: $\sin x$ is strictly increasing on this interval.

Surjective: it still covers $[-1, 1]$.

Hence: f is bijective on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and has inverse $f^{-1}(y) = \arcsin y$.

Lesson: Sometimes, to get injectivity, we must "trim" the domain.

Summary: Domain and Codomain Shape the Function's Nature

Key Insights:

The same formula $f(x)$ can be injective or not, surjective or not — depending on the sets involved.

Domain affects injectivity (how inputs behave).

Codomain affects surjectivity (which outputs we expect).

Restricting or enlarging either can change the nature of f.

Examples Recap:

$f(x) = x^2$: Injective on $[0, \infty)$, Surjective onto $[0, \infty)$.

$f(x) = 2x + 3$: Bijective from $\mathbb{R} \to \mathbb{R}$.

$f(x) = \sin x$: Bijective only when restricted to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Moral: Mathematical properties don't live in formulas — they live in the relationship between sets.